Importando librerias

In [1]:
```
import pandas as pd
import numpy as np
import sqlite3
import matplotlib.pyplot as plt
import seaborn as sns
import warnings

warnings.filterwarnings('ignore')

df_nuevo = pd.read_csv('jobs_cleaned_table.csv', sep=';',encoding='latin1')
conn = sqlite3.connect('jobs.db')
df_nuevo.to_sql('jobs_cleaned_table', conn, if_exists='replace', index=False)
conn.close()
```

Cargando Dataset

In [2]:
```
conn = sqlite3.connect('jobs.db')

query = "SELECT * FROM jobs_cleaned_table WHERE (job_group='Data Scientist' OR j
df = pd.read_sql_query(query,conn)

df['date_posted'] = pd.to_datetime(df['date_posted'].str.split().str[0], errors=

conn.close()
```

EDA inicial

In [3]:
```
df.head()
```

Out[3]:

	site	title	company	date_posted	job_type	skills	level	job_
0	indeed	Business Intelligence Developer / Analyst - Re...	Bluelight Consulting	2026-08-03	fulltime	CSS, JavaScript, Agile, React, SSIS, SQL, DAX,...	Mid-Level	Bu Intelli
1	indeed	Senior Business Intelligence	Bluelight Consulting	2026-08-03	fulltime	Python, SQL, Looker, Excel, Power BI	Senior	Bu Intelli
2	indeed	Senior Data Engineer (Celonis / Microsoft Fabr...	Bertoni Solutions	2026-06-03	fulltime	SQL, ERP, Fabric, Power BI, , Azure	Senior	En
3	indeed	3PL Logistics Data Analyst (Remote) | LATAM	HireHawk	2026-05-03	fulltime	Excel,SQL,Power BI,Data Visualization	Mid-Level	A
4	indeed	SSC Master Data / Pre-Qualification Analyst	Millicom	2026-05-03	fulltime	Agile, Excel, Power BI, , Master	Mid-Level	A

In [4]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 322 entries, 0 to 321
Data columns (total 17 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   site                   322 non-null    object
 1   title                  322 non-null    object
 2   company                322 non-null    object
 3   date_posted            195 non-null    object
 4   job_type               322 non-null    object
 5   skills                 322 non-null    object
 6   level                  322 non-null    object
 7   job_group              322 non-null    object
 8   workplace              322 non-null    object
 9   city                   322 non-null    object
 10  state                  322 non-null    object
 11  country                322 non-null    object
 12  city_state             322 non-null    object
 13  experience             322 non-null    int64
 14  education              322 non-null    object
 15  programming_languages  151 non-null    object
 16  languages              322 non-null    object
dtypes: int64(1), object(16)
memory usage: 42.9+ KB
```

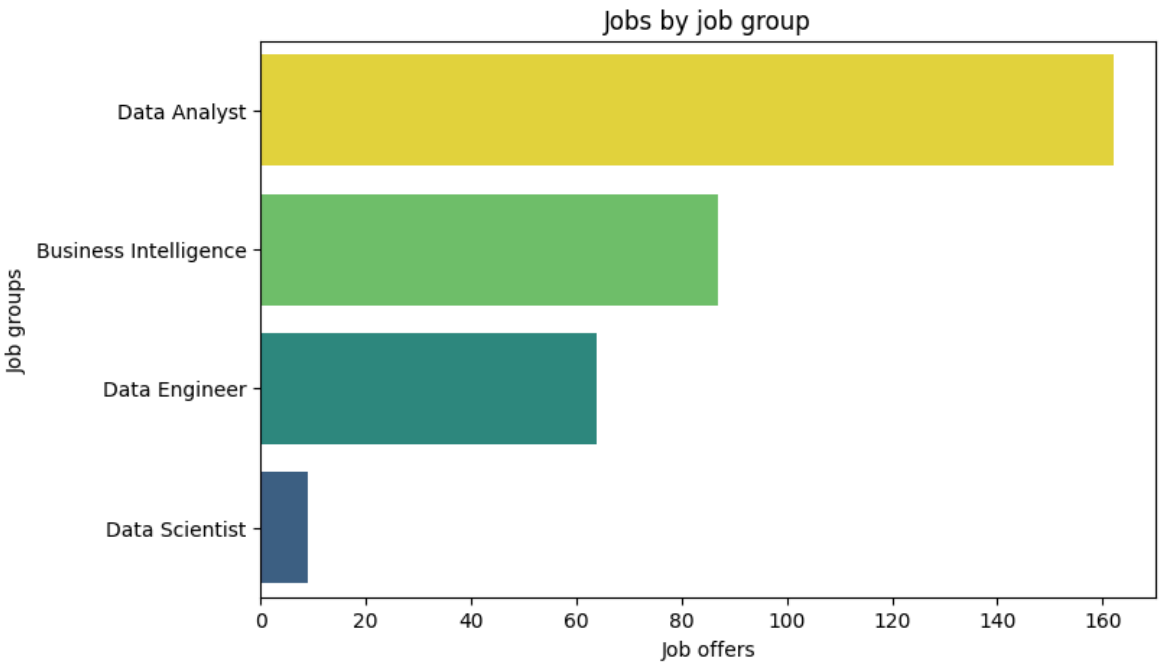
In [5]:
```
df.describe(include='all')
```

Out[5]:

	site	title	company	date_posted	job_type	skills	level	job_
count	322	322	322	195	322	322	322	
unique	2	253	174	24	1	44	4	
top	linkedin	Data Engineer	Inetum	2026-12-03	fulltime	SQL,Excel,Power BI,Data Visualization	Mid-Level	A
freq	185	8	16	25	322	51	225	
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
75%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
max	NaN	NaN	NaN	NaN	NaN	NaN	NaN	

In [6]:

```
job_groups_counts = df['job_group'].value_counts()

def plot_bars(x,y,figsize=(8,5),title=None,xlabel=(None),ylabel=(None),palette=N

    plt.figure(figsize=figsize)
    cmap = plt.cm.get_cmap(palette).reversed()
    palette = [cmap(x) for x in np.linspace(0,0.7,len(job_groups_counts))]

    sns.barplot(x=x,y=y,palette=palette)

    plt.title(title)
    plt.xlabel(xlabel)
    plt.ylabel(ylabel)
    plt.show()

plot_bars(x=job_groups_counts.values,y=job_groups_counts.index,title='Jobs by jo
```



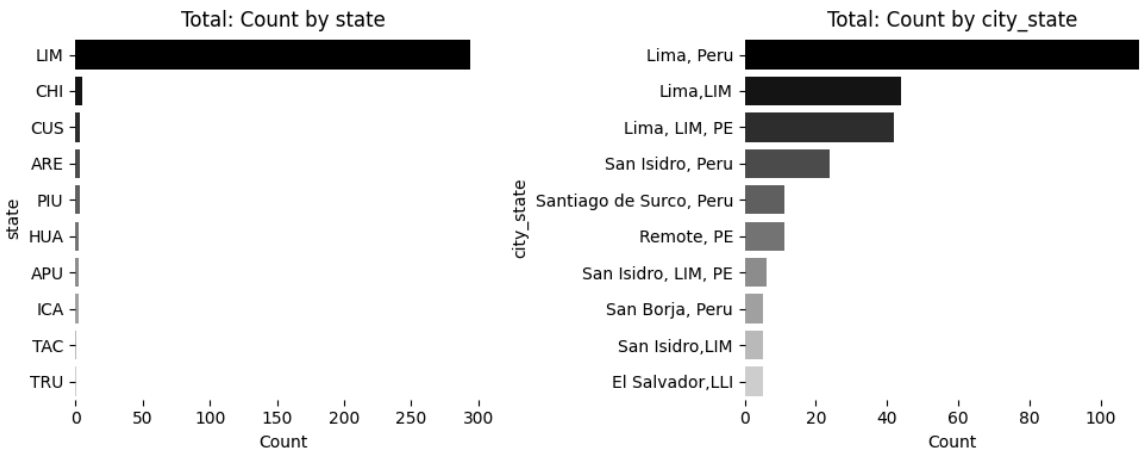
Análisis exploratorio de datos

Lugar

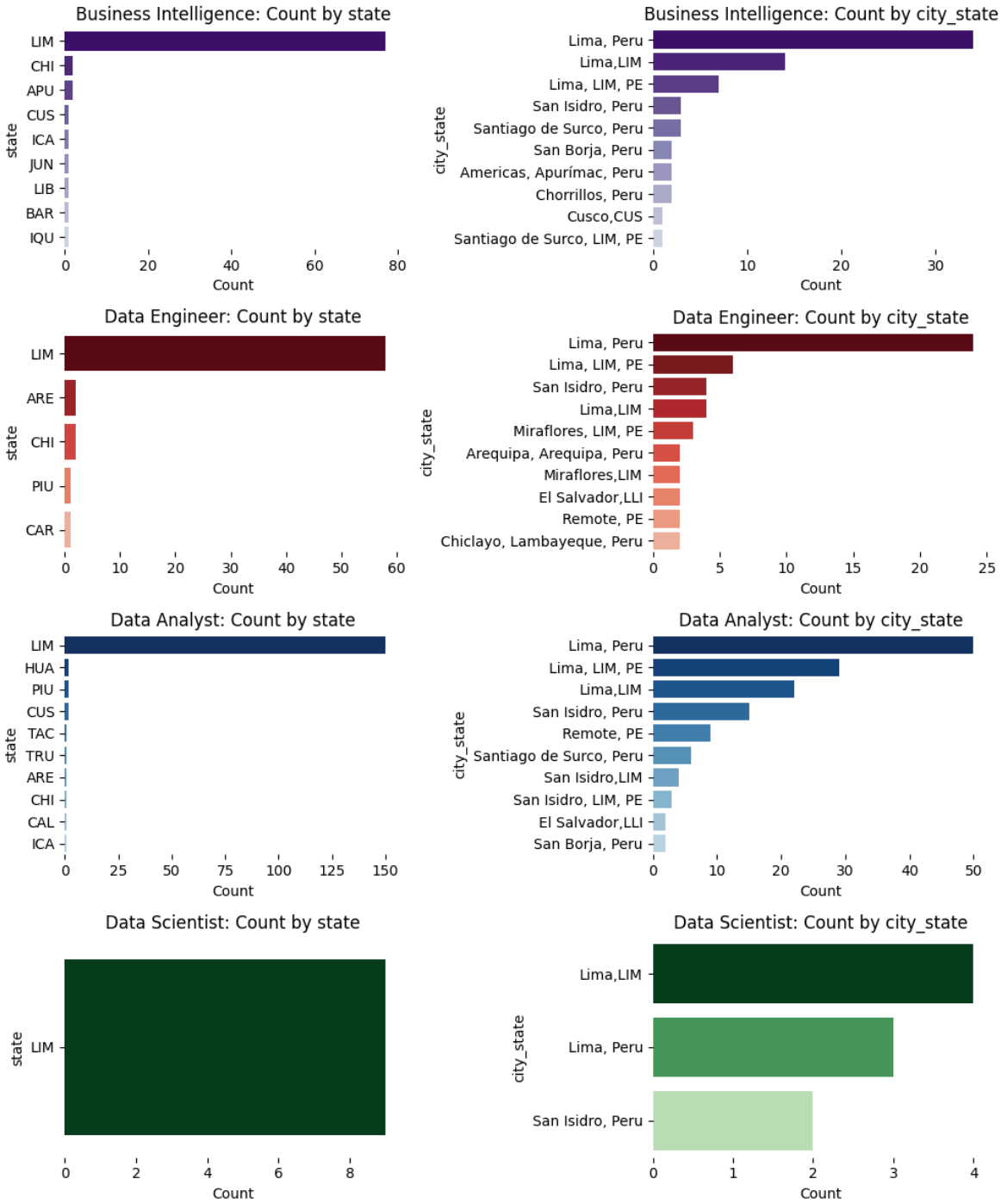
In [7]:
```
import sqlite3

def plot_multiple_bars(columns,category=None,figsize=(10,12),count_percentage=Fa

    rows = df[category].unique() if category is not None else ['Total']
    n_rows, n_cols = len(rows), len(columns)

    fig, axes = plt.subplots(n_rows,n_cols,figsize=figsize,squeeze=False)

    category_palettes = {
        'Data Analyst': 'Blues',
        'Data Engineer': 'Reds',
        'Data Scientist': 'Greens',
        'Business Intelligence': 'Purples',
    }

    for i, row in enumerate(rows):
        group_data = df[df[category] == row] if category is not None else df

        for j, column in enumerate(columns):

            if aggregate_column is None:
                if top_10 and list_values is False:
                    column_top = group_data[column].value_counts().head(10)

                elif list_values is not False and top_10 is None:
                    column_top=(group_data[column].str.split(', ').explode()
                                .value_counts()
                                .dropna()
                                .sort_values(ascending=False))
                    column_top = column_top.drop(labels='',errors='ignore')
```

```
            elif list_values is not False and top_10 is not None:
                column_top=(group_data[column].str.split(', ').explode()
                            .value_counts()
                            .dropna()
                            .sort_values(ascending=False).head(10))
                column_top = column_top.drop(labels='',errors='ignore')

            else:
                column_top = group_data[column].value_counts()

        else:
            grouped =(
                group_data.explode(column)
                .groupby(column)[aggregate_column]
                .mean()
                .sort_values(ascending=False)
            )

            if list_values:
                group_data = group_data.reset_index(drop=True)
                grouped = pd.DataFrame({
                    column: group_data[column].str.split(', ').explode(),
                    aggregate_column: group_data[aggregate_column]
                })
                grouped = grouped.groupby(column)[aggregate_column].mean().s

            if '' in grouped.index:
                grouped = grouped.drop('')

            column_top = grouped.head(10) if top_10 else grouped

        ax = axes[i,j]
        total = group_data[column].count()

        for spine in ax.spines.values():
            spine.set_visible(False)

        if category is not None:
            palette = category_palettes.get(row,'Blues')
            cmap = plt.cm.get_cmap(palette).reversed()
            palette = [cmap(x) for x in np.linspace(0,0.7,len(column_top))]

        else:
            cmap = plt.cm.get_cmap('Greys').reversed()
            palette = [cmap(x) for x in np.linspace(0,0.7,len(column_top))]

        kwargs ={
            'y':column_top.values,
            'x':column_top.index
        } if x_y else {
            'x':column_top.values,
            'y':column_top.index
        }

        sns.barplot(ax=ax,palette=palette, **kwargs)

        aggregate_label = 'Count' if aggregate_column is None else f'Avg{agg
        ax.set_title(f'{row}: {aggregate_label} by {column}')
        ax.set_xlabel(aggregate_label if not x_y else column)
```

```
            ax.set_ylabel(column if not x_y else aggregate_label)

            if count_percentage:
                for k, value in enumerate(column_top.values):
                    percentage = value/total*100
                    annotation_kwargs = {
                        'text': f'{value:.1f}({percentage:.1f}%)',
                        'xy':(k, value) if x_y else (value,k),
                        'xytext': (-30,5) if x_y else (5,0),
                        'textcoords': 'offset points',
                        'va':'center',
                        'ha':'left',
                        'fontsize':10,
                        'color':'black'
                    }
                    ax.annotate(**annotation_kwargs)

    plt.tight_layout()
    plt.show()

columns_location = ['state','city_state']

plot_multiple_bars(columns_location,figsize=(10,4),top_10=True)
```

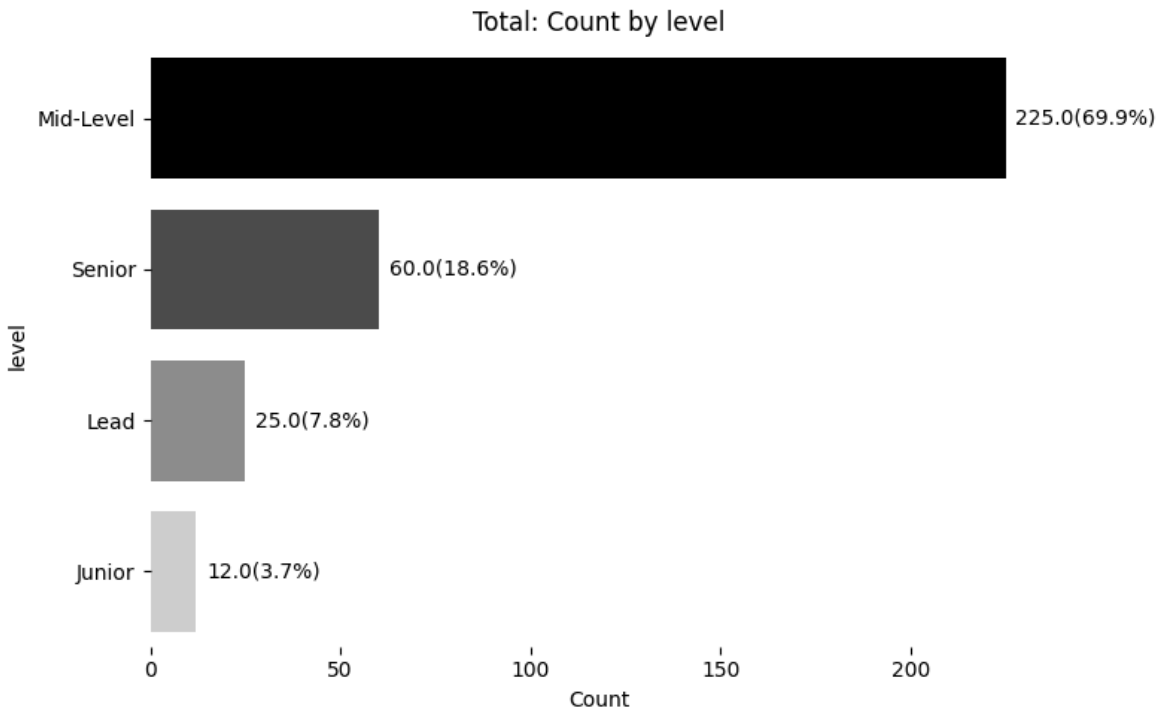


In [8]:
```
plot_multiple_bars(columns_location,category='job_group',figsize=(10,12),top_10=
```

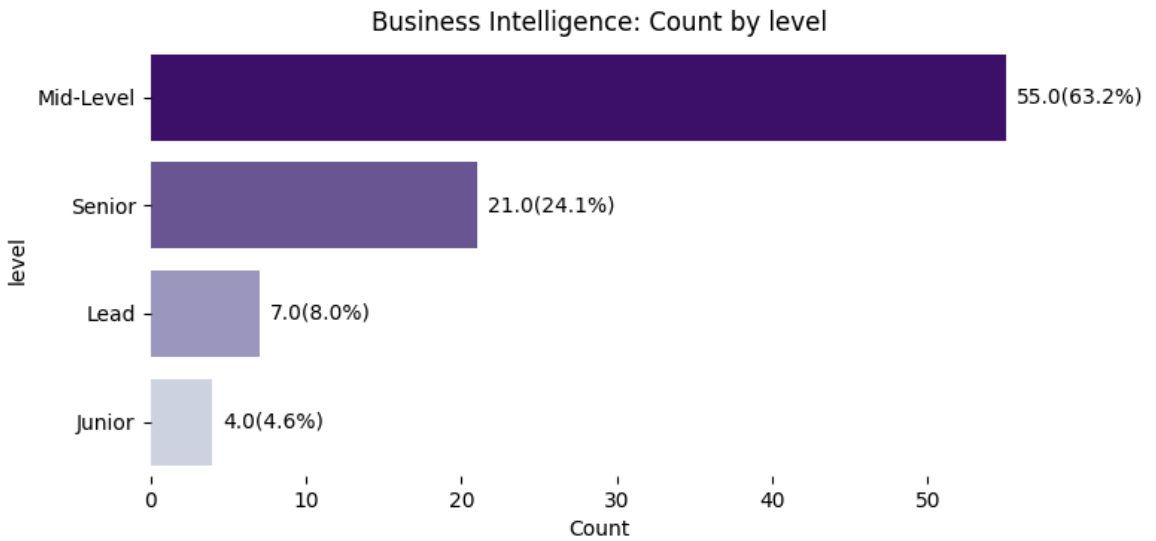


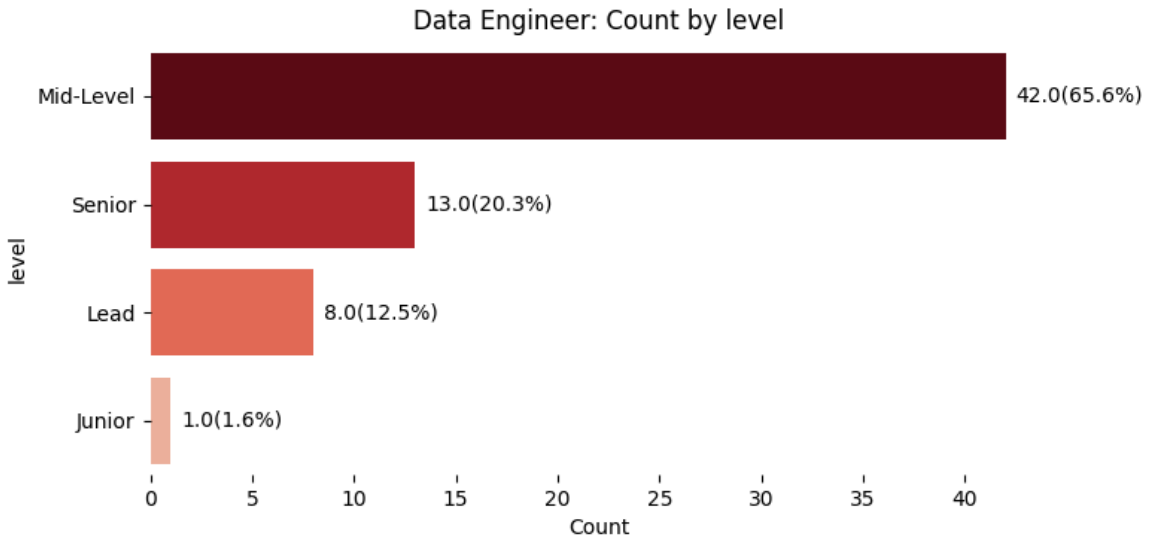
Niveles

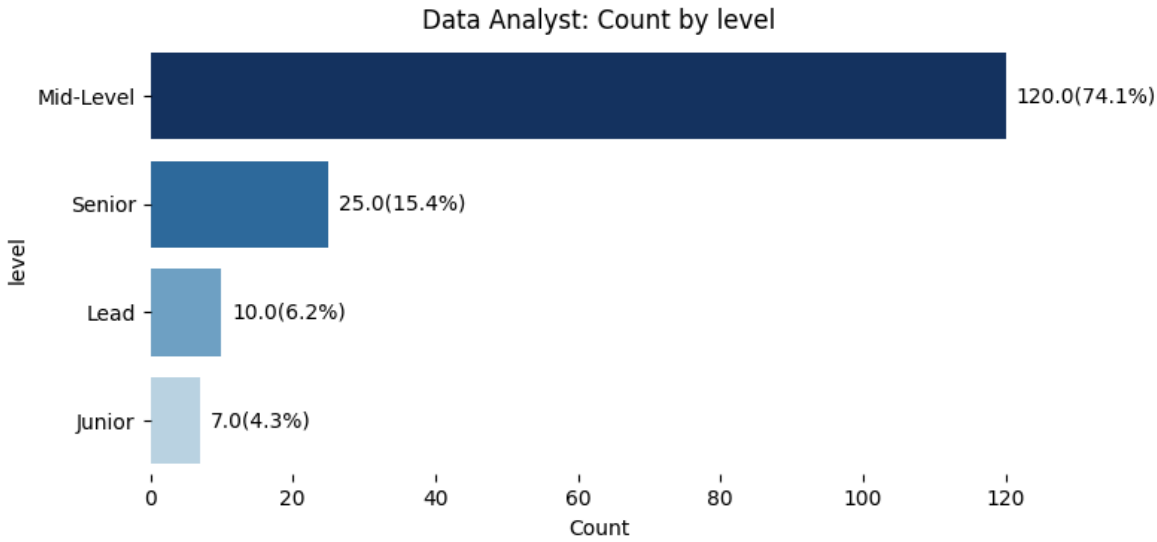
In [9]:
```
columns_level = ('level',)

plot_multiple_bars(columns_level,figsize=(8,5),count_percentage=True)
```

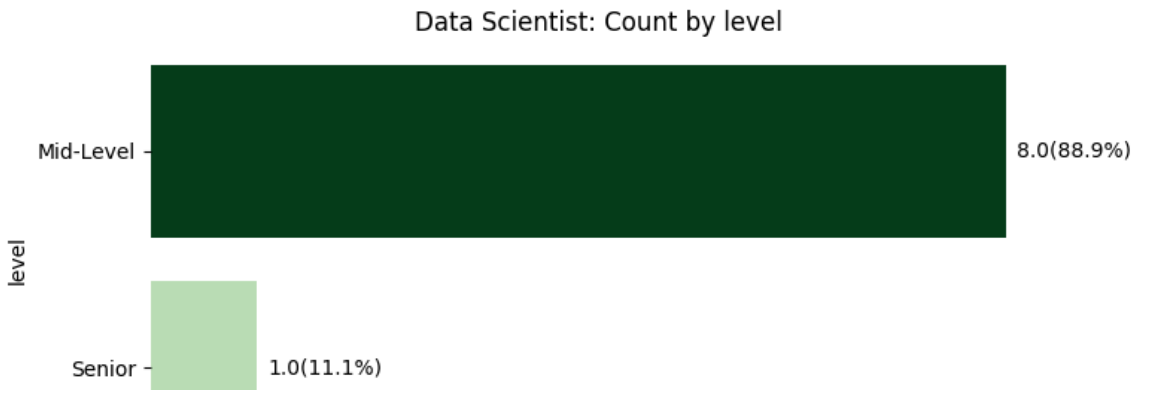


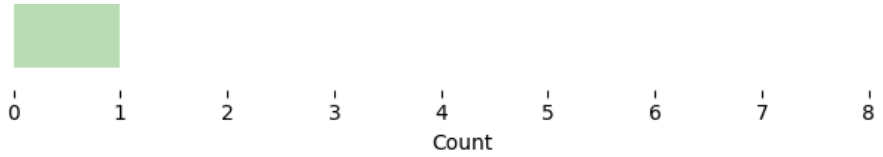
In [10]:
```
plot_multiple_bars(columns_level,category='job_group',figsize=(8,15),count_perce
```


Business Intelligence: Count by level
Mid-Level
Senior
Lead
Junior
level
55.0(63.2%)
21.0(24.1%)
7.0(8.0%)
4.0(4.6%)
0
10
20
30
40
50
Count


Data Engineer: Count by level
Mid-Level
Senior
Lead
Junior
level
42.0(65.6%)
13.0(20.3%)
8.0(12.5%)
1.0(1.6%)
0
5
10
15
20
25
30
35
40
Count

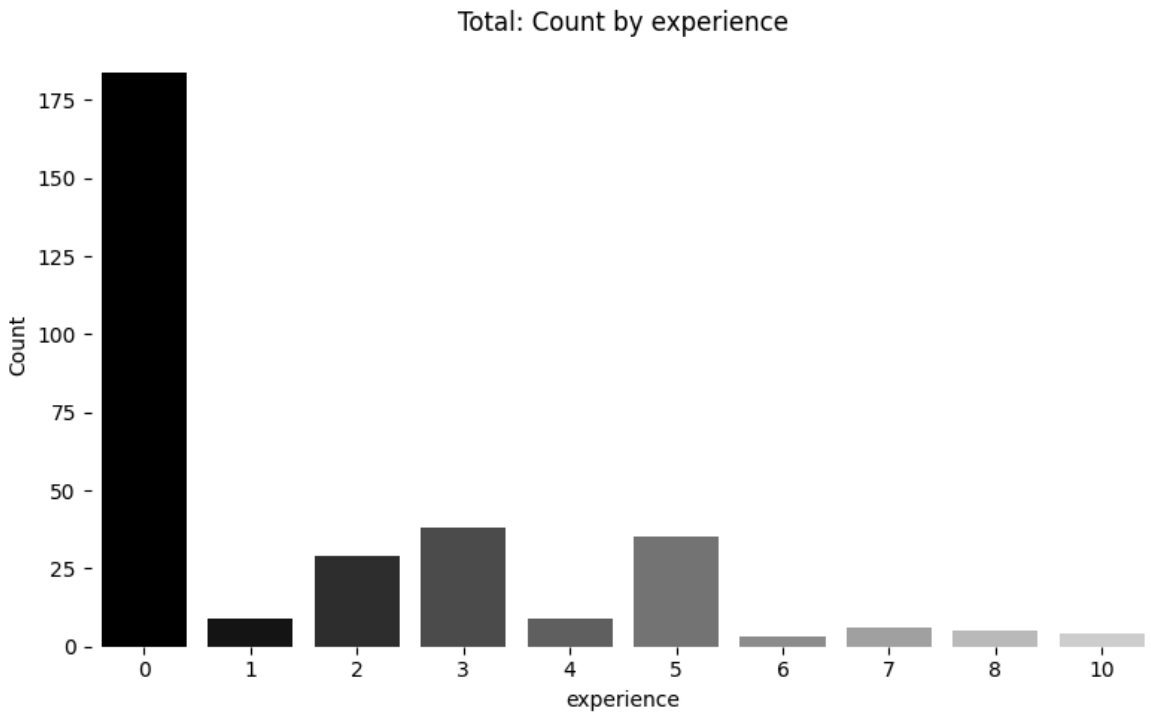

Data Analyst: Count by level
Mid-Level
Senior
Lead
Junior
level
120.0(74.1%)
25.0(15.4%)
10.0(6.2%)
7.0(4.3%)
0
20
40
60
80
100
120
Count


Data Scientist: Count by level
Mid-Level
Senior
level
8.0(88.9%)
1.0(11.1%)



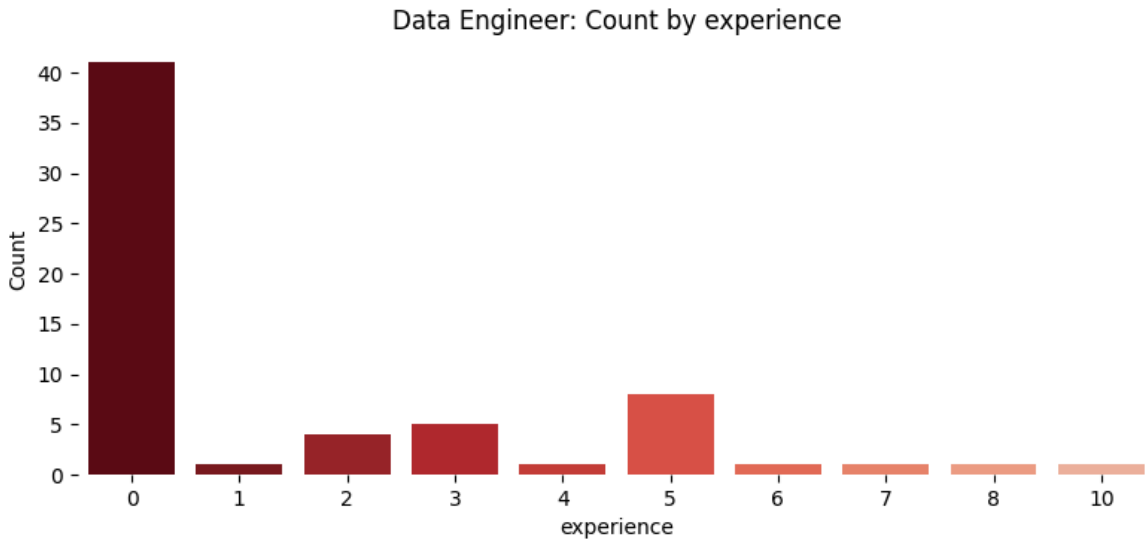
Experiencia

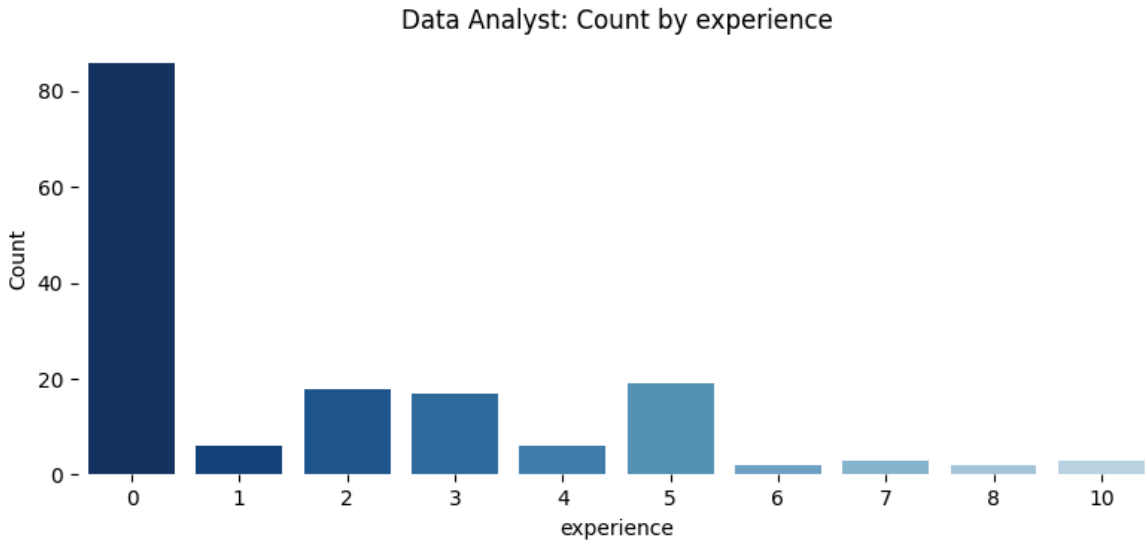
In [11]:
```
columns_experience = ('experience',)

plot_multiple_bars(columns_experience,figsize=(8,5),x_y=True)
```

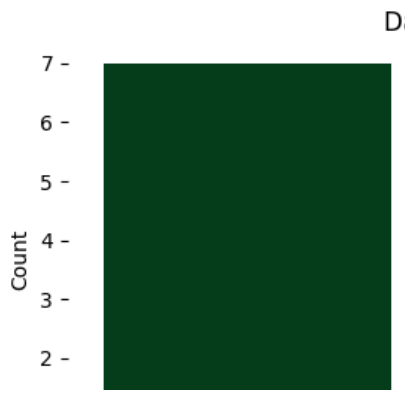


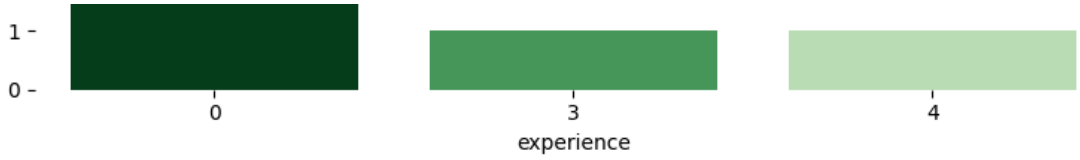
In [12]:
```
plot_multiple_bars(columns_experience,category='job_group',figsize=(8,15),x_y=Tr
```

Business Intelligence: Count by experience
Count
experience
0
1
2
3
4
5
7
8
0
10
20
30
40
50


Data Engineer: Count by experience
Count
experience
0
1
2
3
4
5
6
7
8
10
0
5
10
15
20
25
30
35
40

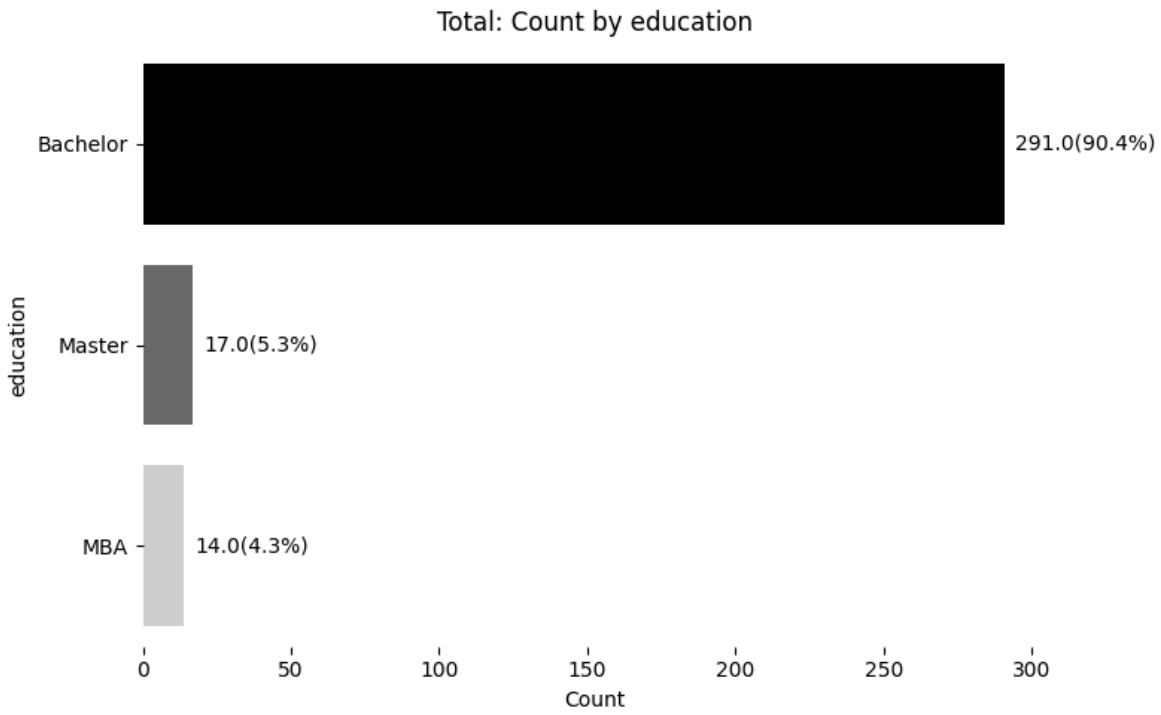

Data Analyst: Count by experience
Count
experience
0
1
2
3
4
5
6
7
8
10
0
20
40
60
80


Data Scientist: Count by experience
Count
2
3
4
5
6
7

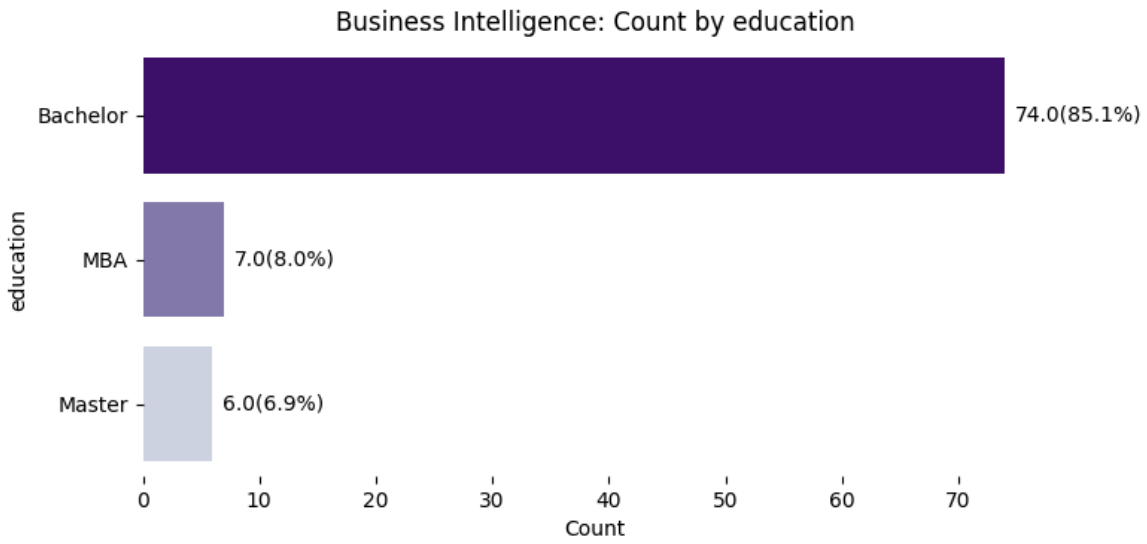


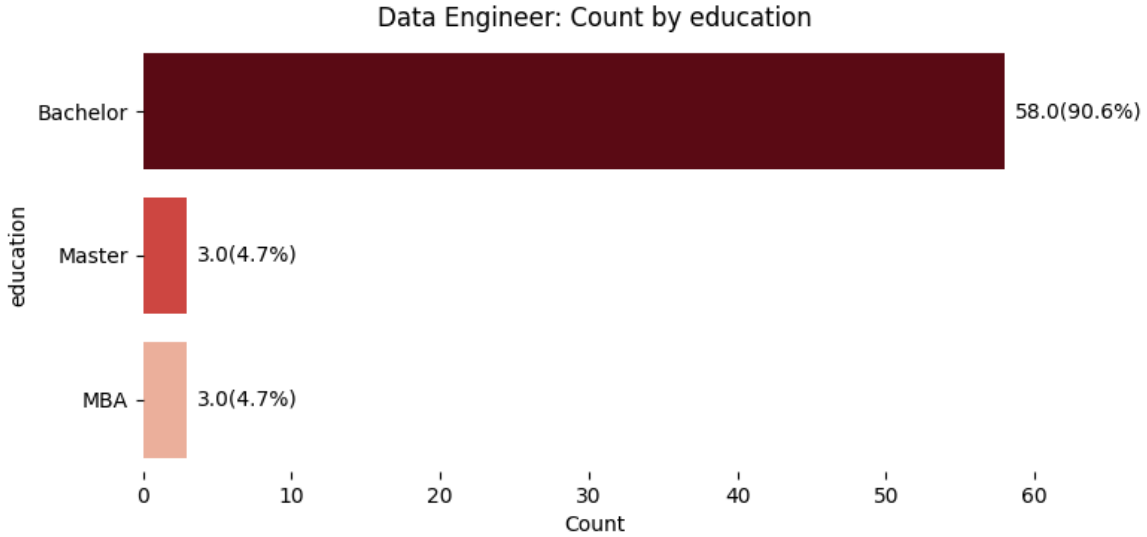
Grado de educación

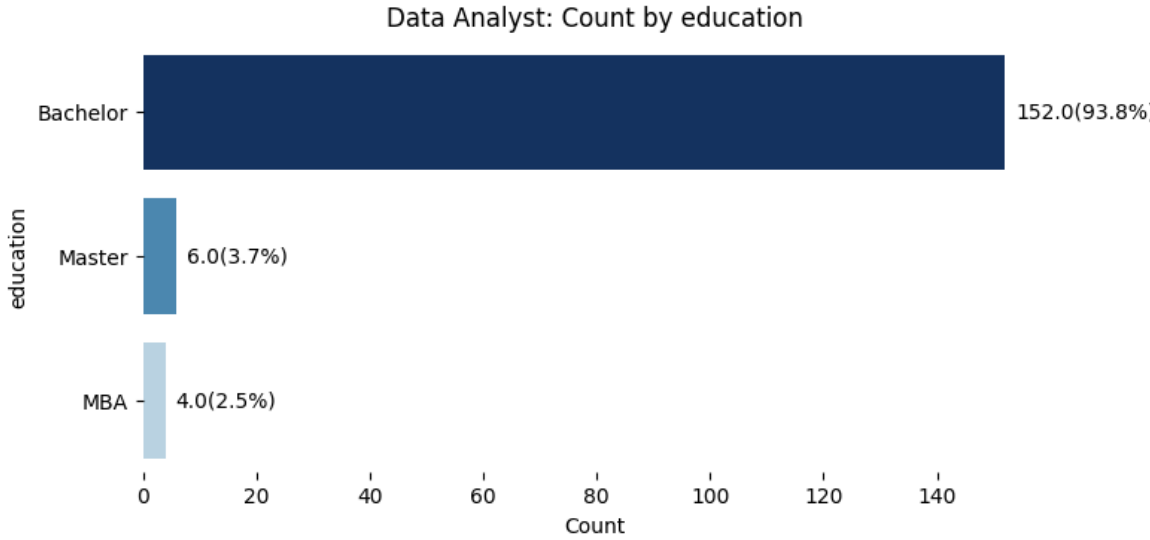
In [13]:
```
columns_education = ('education',)

plot_multiple_bars(columns_education,figsize=(8,5),count_percentage=True)
```

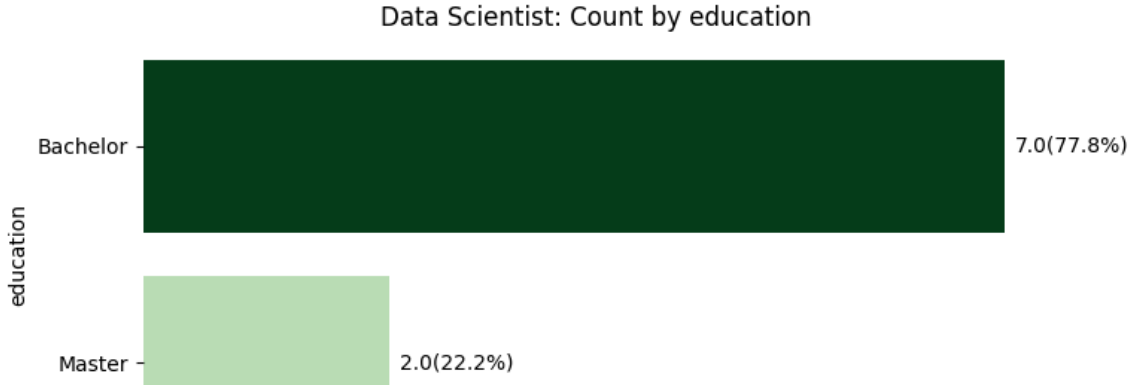


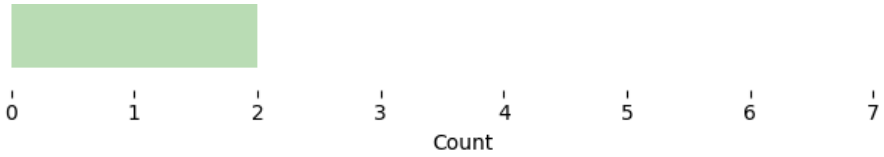
In [14]:
```
plot_multiple_bars(columns_education,category='job_group',figsize=(8,15),count_p
```


Business Intelligence: Count by education
Bachelor
MBA
Master
education
74.0(85.1%)
7.0(8.0%)
6.0(6.9%)
0
10
20
30
40
50
60
70
Count


Data Engineer: Count by education
Bachelor
Master
MBA
education
58.0(90.6%)
3.0(4.7%)
3.0(4.7%)
0
10
20
30
40
50
60
Count

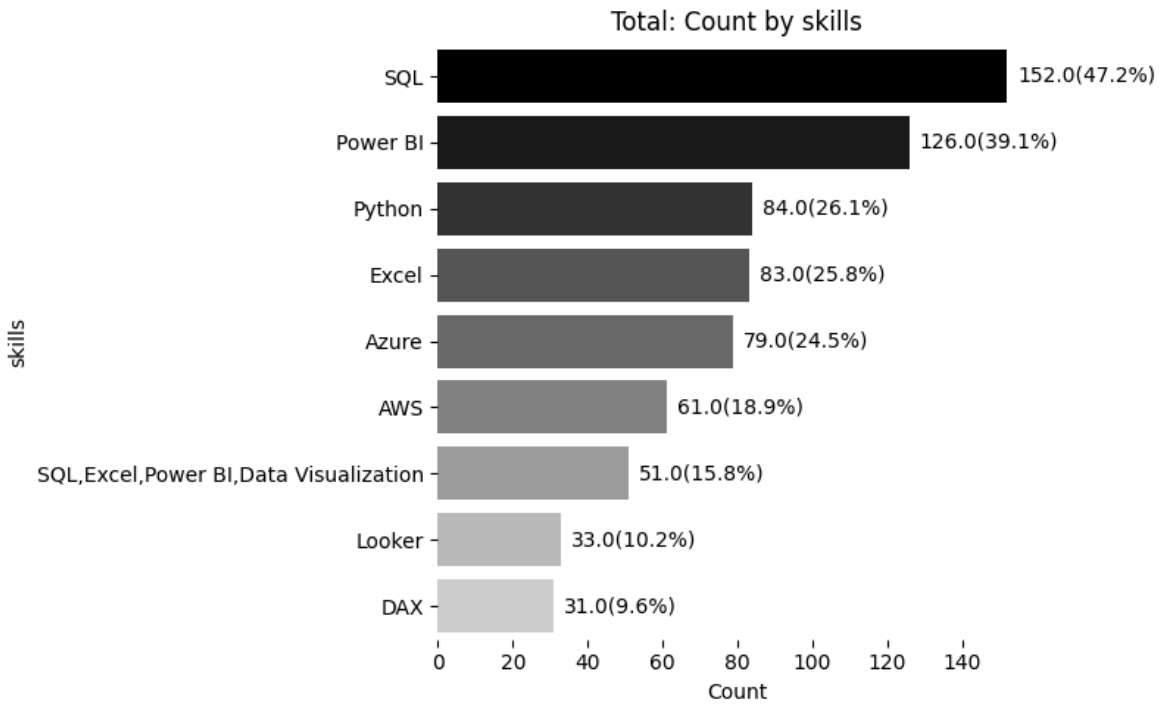

Data Analyst: Count by education
Bachelor
Master
MBA
education
152.0(93.8%)
6.0(3.7%)
4.0(2.5%)
0
20
40
60
80
100
120
140
Count


Data Scientist: Count by education
Bachelor
Master
education
7.0(77.8%)
2.0(22.2%)

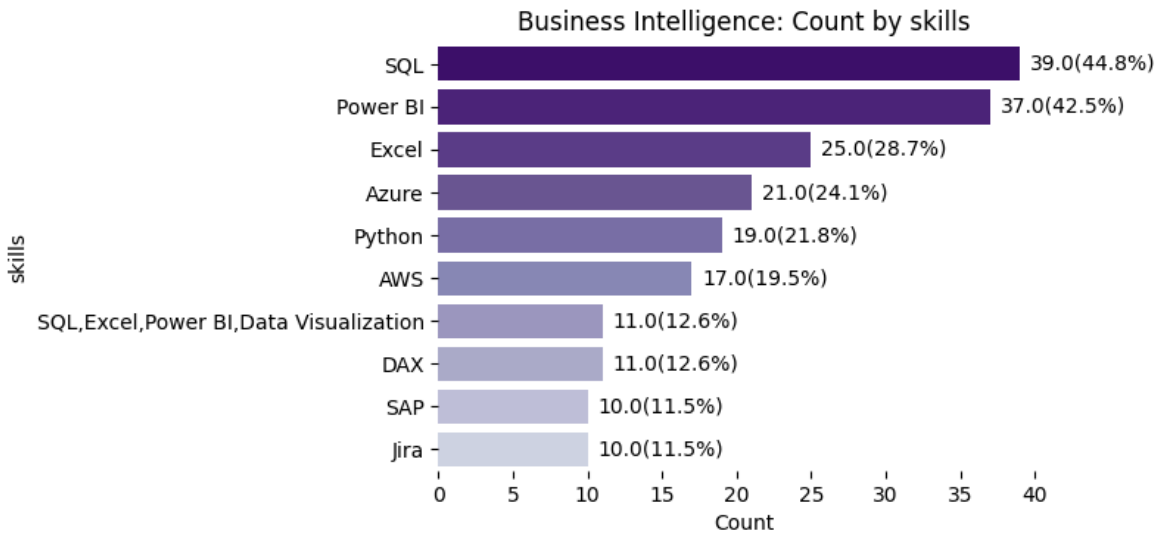


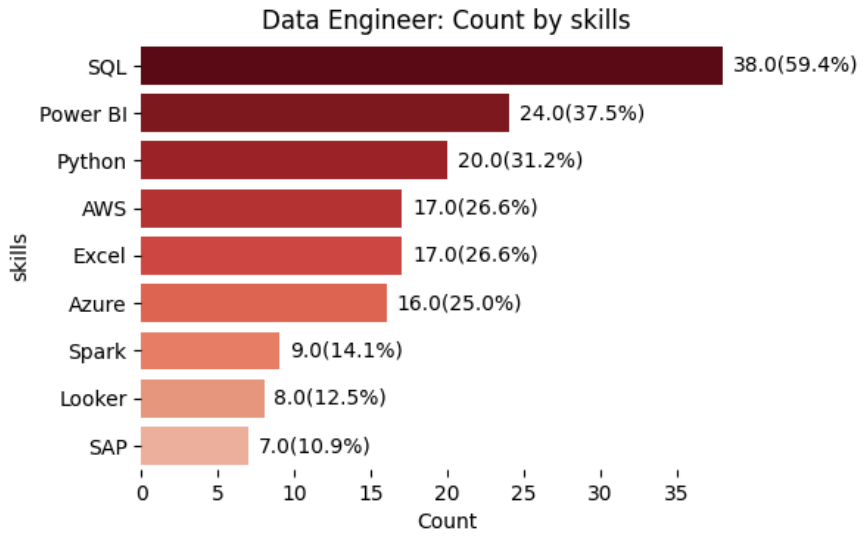
Habilidades

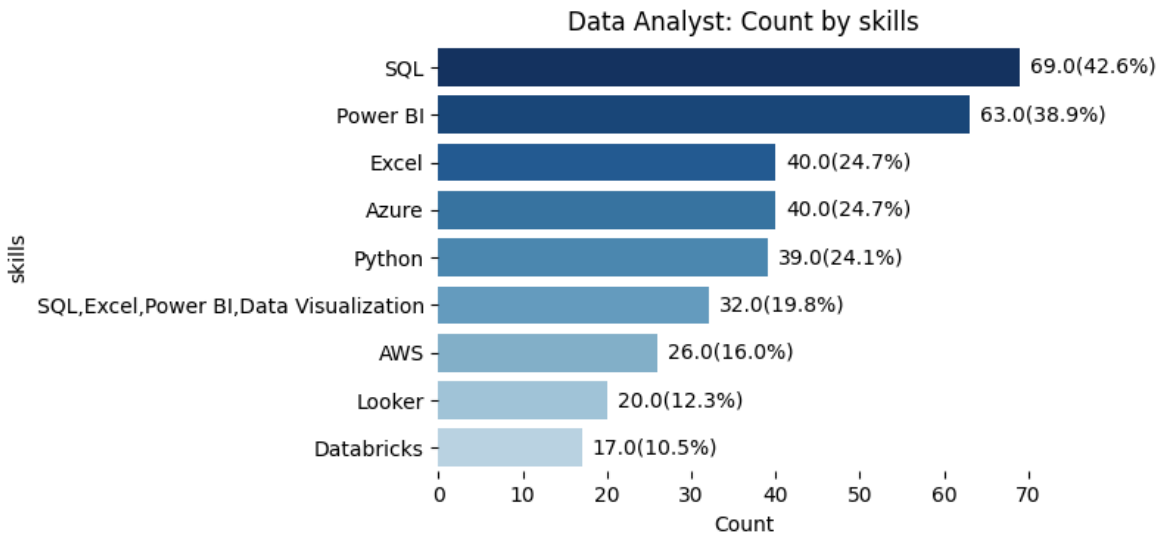
In [15]:
```
columns_skills = ['skills',]

plot_multiple_bars(columns_skills,figsize=(8,5),list_values=True,count_percentag
```

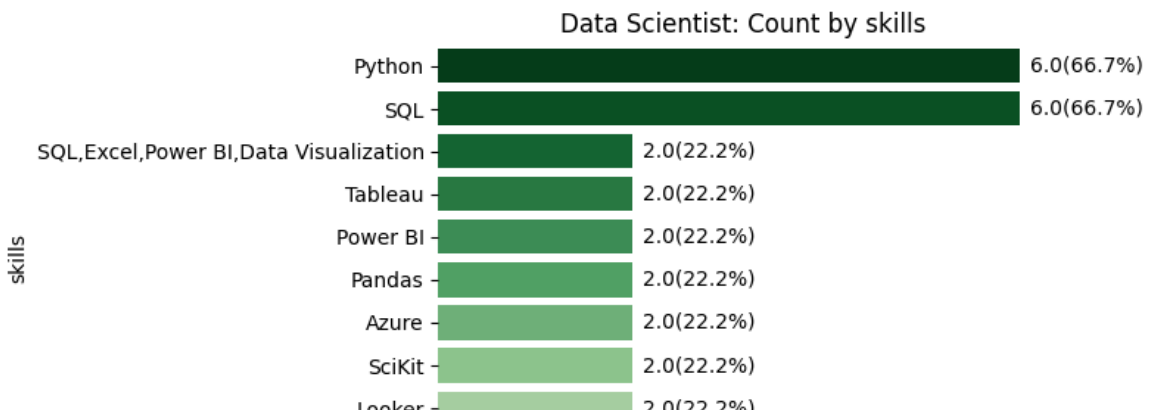


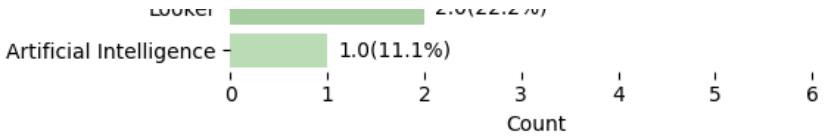
In [16]:
```
plot_multiple_bars(columns_skills,category='job_group',figsize=(8,15),list_value
```


Business Intelligence: Count by skills
SQL
Power BI
Excel
Azure
Python
AWS
SQL,Excel,Power BI,Data Visualization
DAX
SAP
Jira
39.0(44.8%)
37.0(42.5%)
25.0(28.7%)
21.0(24.1%)
19.0(21.8%)
17.0(19.5%)
11.0(12.6%)
11.0(12.6%)
10.0(11.5%)
10.0(11.5%)
skills
0 5 10 15 20 25 30 35 40
Count


Data Engineer: Count by skills
SQL
Power BI
Python
AWS
Excel
Azure
Spark
Looker
SAP
38.0(59.4%)
24.0(37.5%)
20.0(31.2%)
17.0(26.6%)
17.0(26.6%)
16.0(25.0%)
9.0(14.1%)
8.0(12.5%)
7.0(10.9%)
skills
0 5 10 15 20 25 30 35
Count

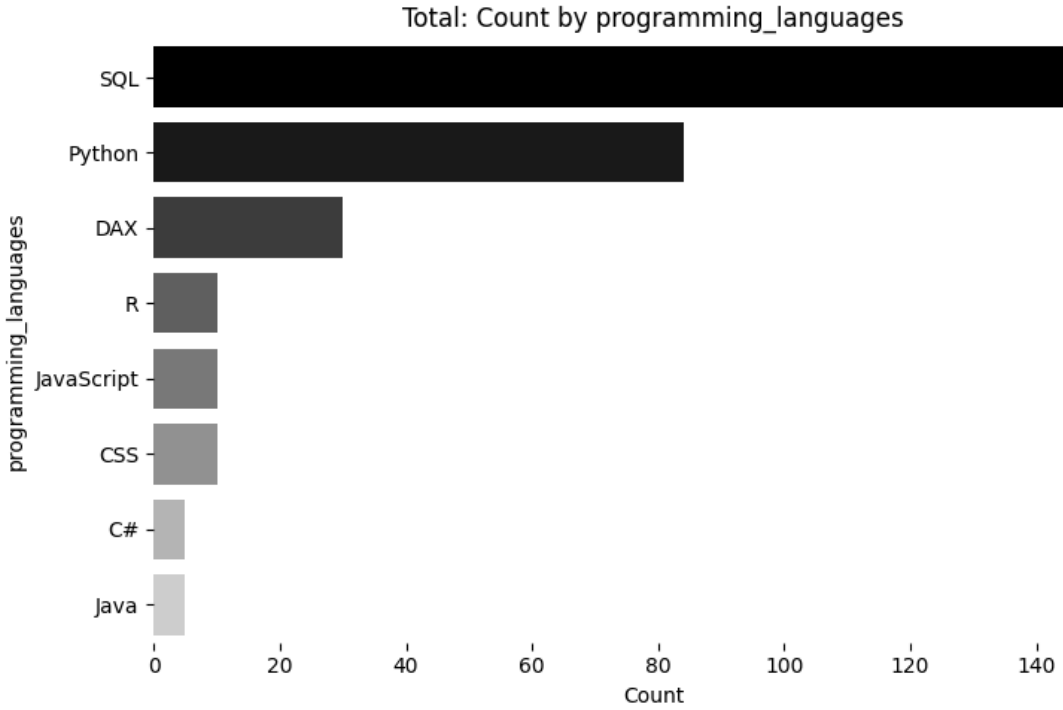

Data Analyst: Count by skills
SQL
Power BI
Excel
Azure
Python
SQL,Excel,Power BI,Data Visualization
AWS
Looker
Databricks
69.0(42.6%)
63.0(38.9%)
40.0(24.7%)
40.0(24.7%)
39.0(24.1%)
32.0(19.8%)
26.0(16.0%)
20.0(12.3%)
17.0(10.5%)
skills
0 10 20 30 40 50 60 70
Count


Data Scientist: Count by skills
Python
SQL
SQL,Excel,Power BI,Data Visualization
Tableau
Power BI
Pandas
Azure
SciKit
6.0(66.7%)
6.0(66.7%)
2.0(22.2%)
2.0(22.2%)
2.0(22.2%)
2.0(22.2%)
2.0(22.2%)
2.0(22.2%)
skills

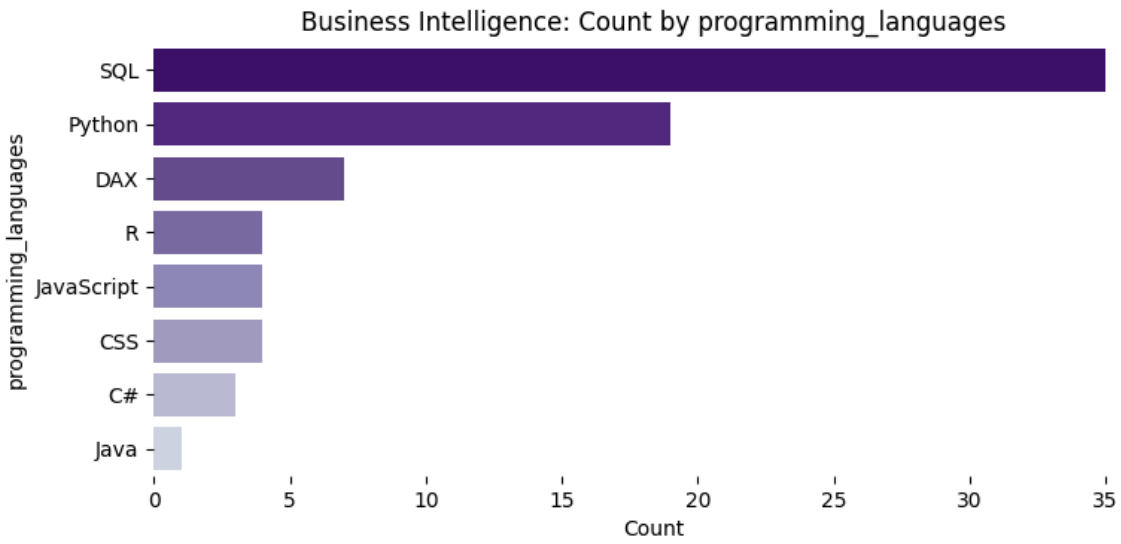


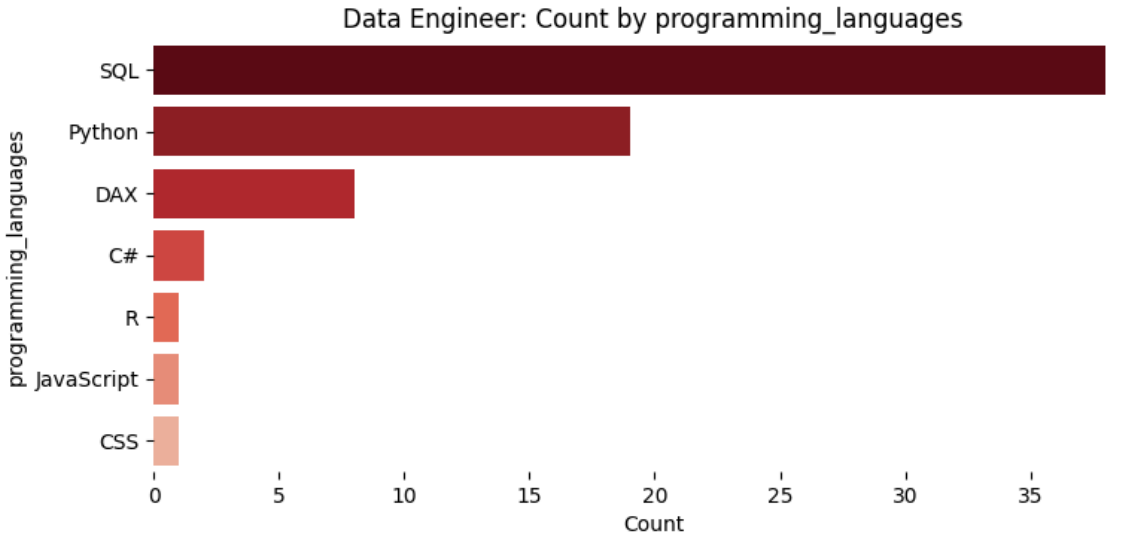
Lenguajes de programación

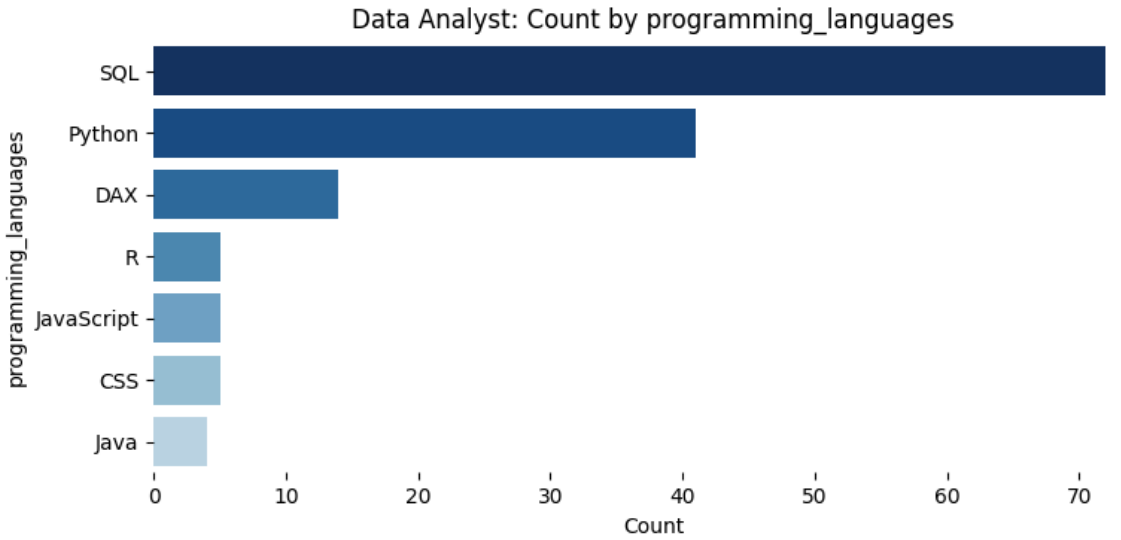
In [17]:
```
column_programming_language = ('programming_languages',)

plot_multiple_bars(column_programming_language,list_values=True,top_10=True,figs
```

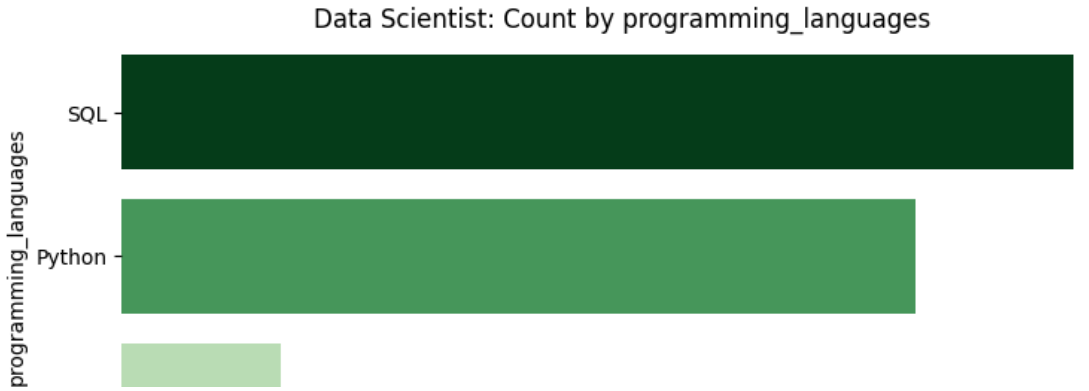


In [18]:
```
plot_multiple_bars(column_programming_language,category='job_group',list_values=
```


Business Intelligence: Count by programming_languages
programming_languages
SQL
Python
DAX
R
JavaScript
CSS
C#
Java
0
5
10
15
20
25
30
35
Count

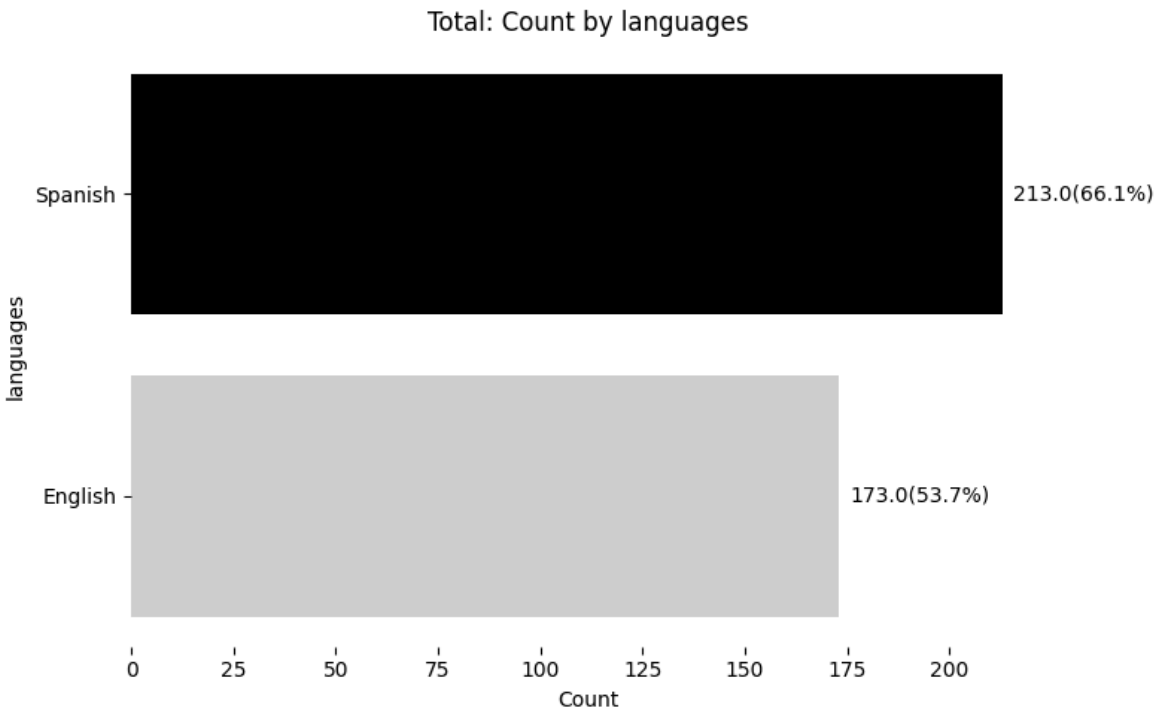

Data Engineer: Count by programming_languages
programming_languages
SQL
Python
DAX
C#
R
JavaScript
CSS
0
5
10
15
20
25
30
35
Count


Data Analyst: Count by programming_languages
programming_languages
SQL
Python
DAX
R
JavaScript
CSS
Java
0
10
20
30
40
50
60
70
Count

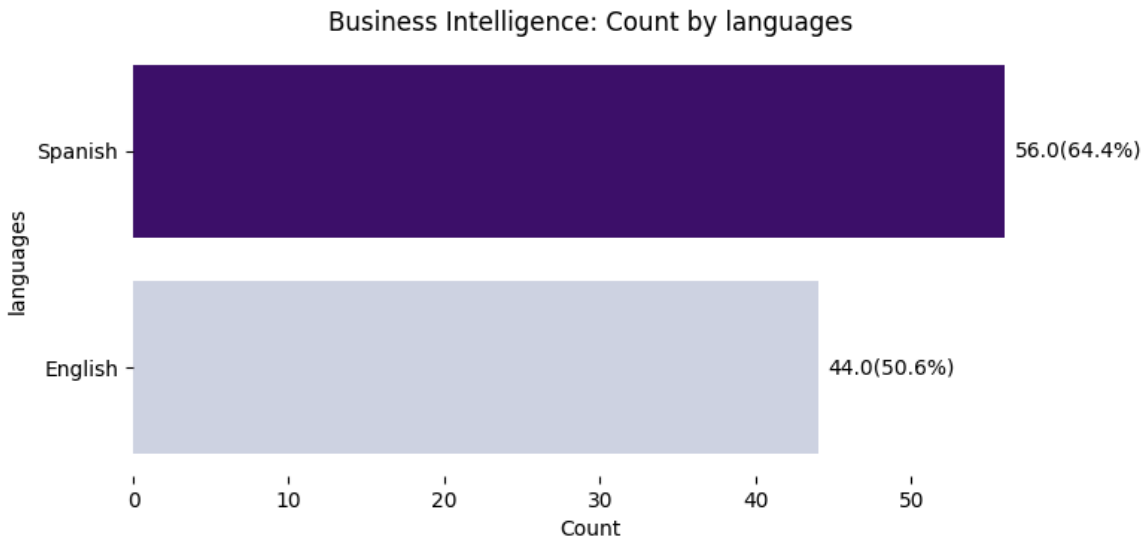

Data Scientist: Count by programming_languages
programming_languages
SQL
Python

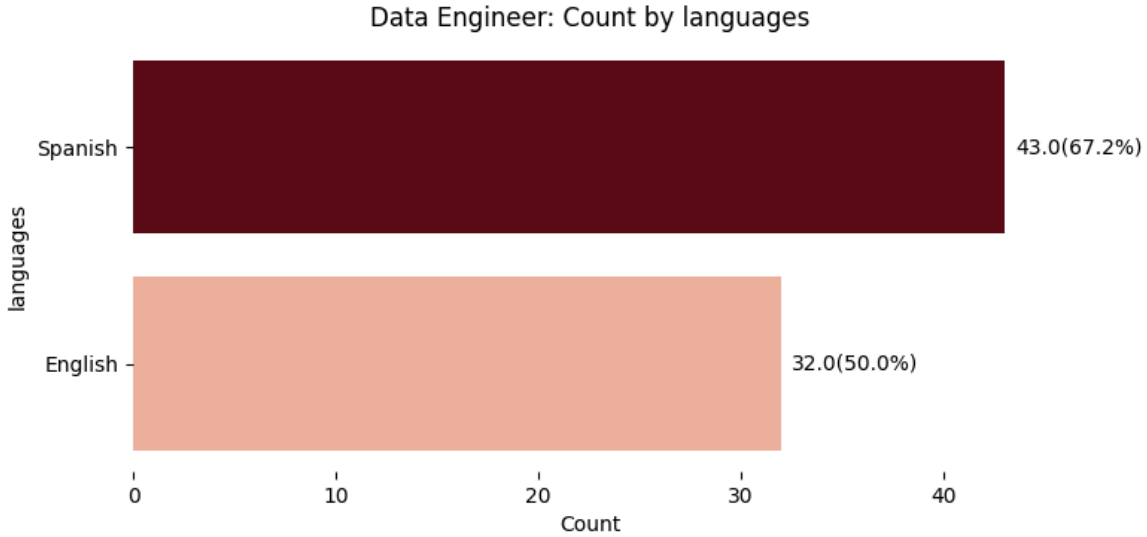
Idiomas mejor pagados

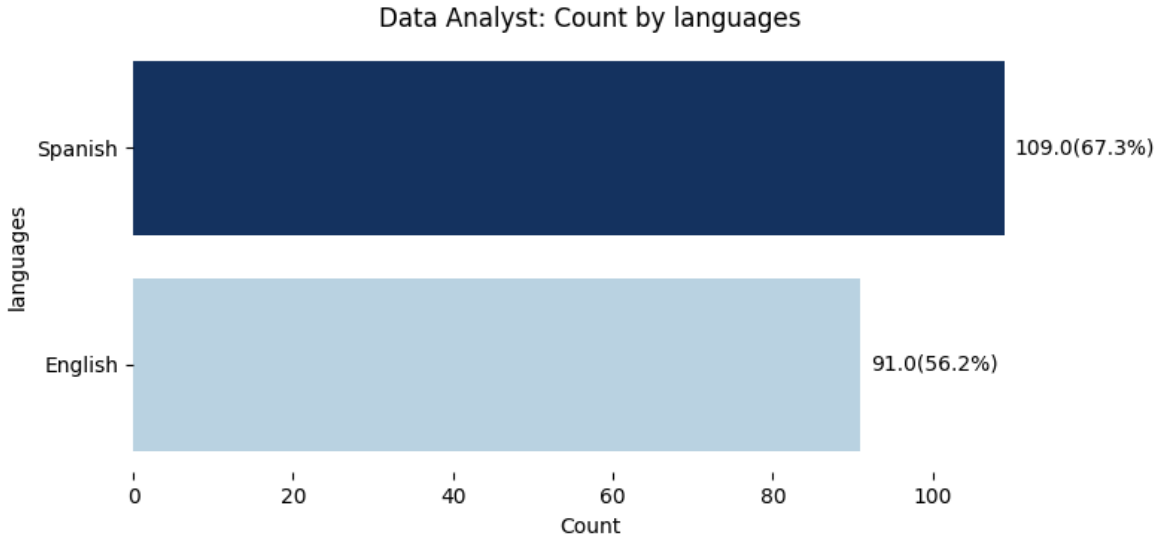
In [19]:
```
column_languages = ('languages',)

plot_multiple_bars(column_languages,list_values=True,figsize=(8,5),count_percent
```

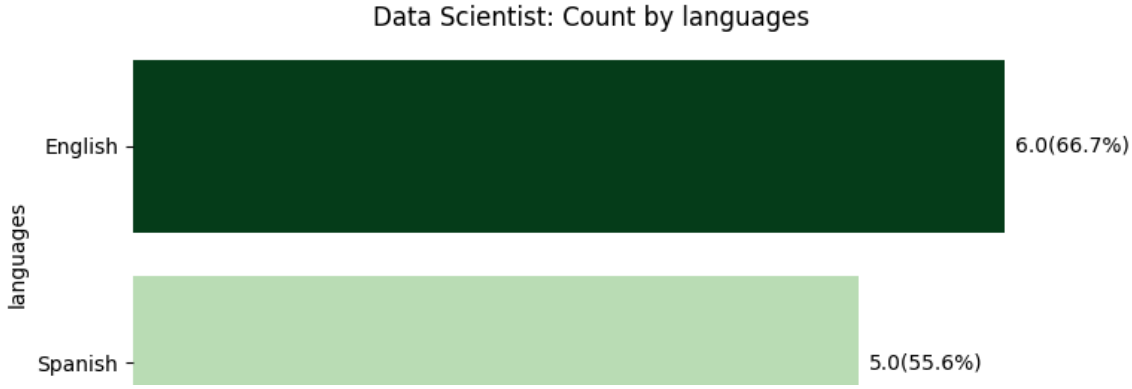


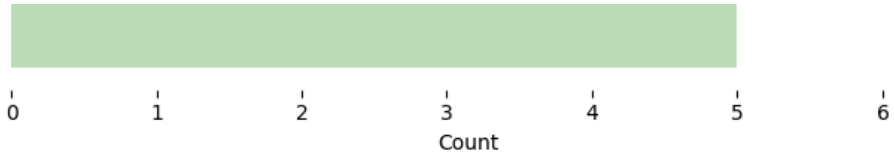
In [20]:
```
plot_multiple_bars(column_languages,category = 'job_group',list_values=True,figs
```


Business Intelligence: Count by languages
Spanish
English
languages
56.0(64.4%)
44.0(50.6%)
0
10
20
30
40
50
Count


Data Engineer: Count by languages
Spanish
English
languages
43.0(67.2%)
32.0(50.0%)
0
10
20
30
40
Count

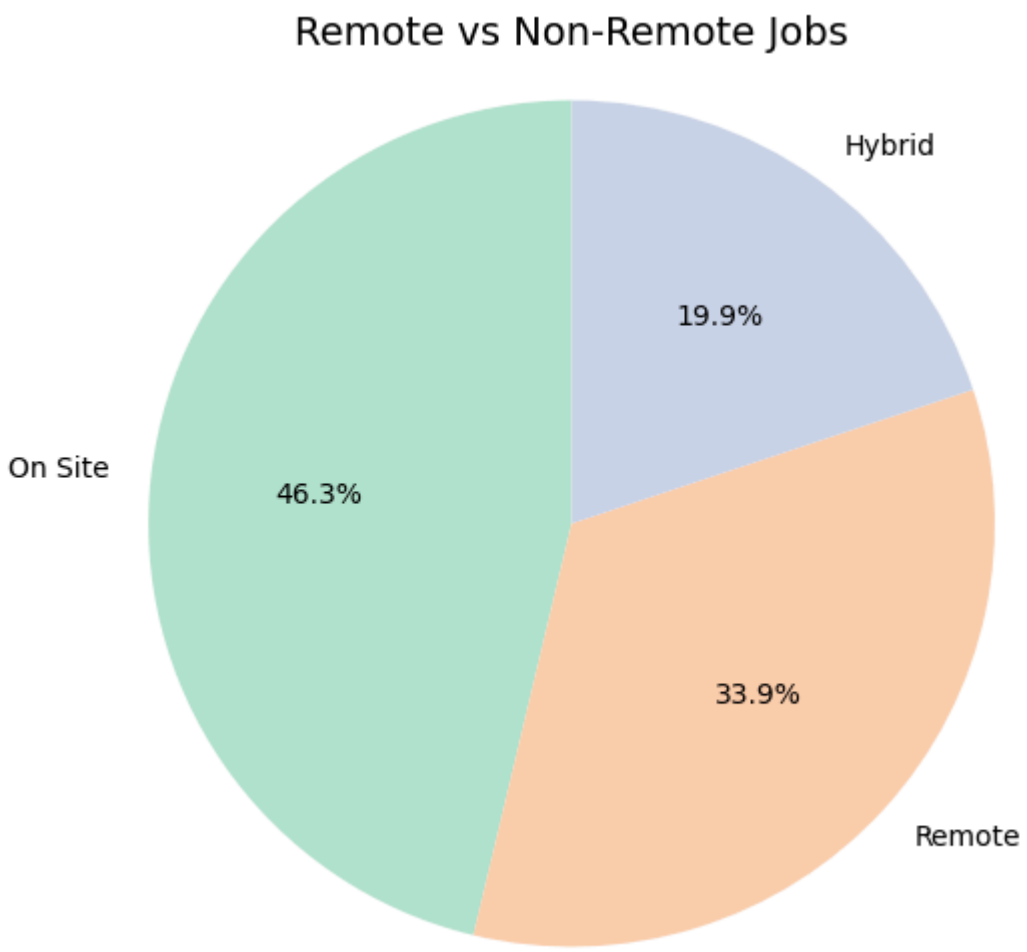

Data Analyst: Count by languages
Spanish
English
languages
109.0(67.3%)
91.0(56.2%)
0
20
40
60
80
100
Count


Data Scientist: Count by languages
English
Spanish
languages
6.0(66.7%)
5.0(55.6%)

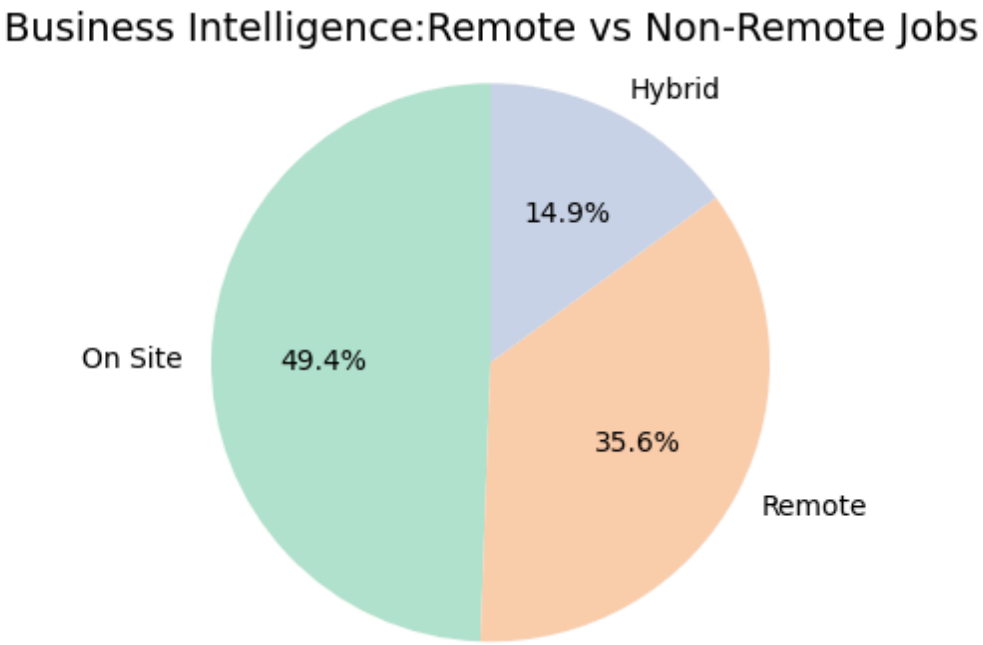


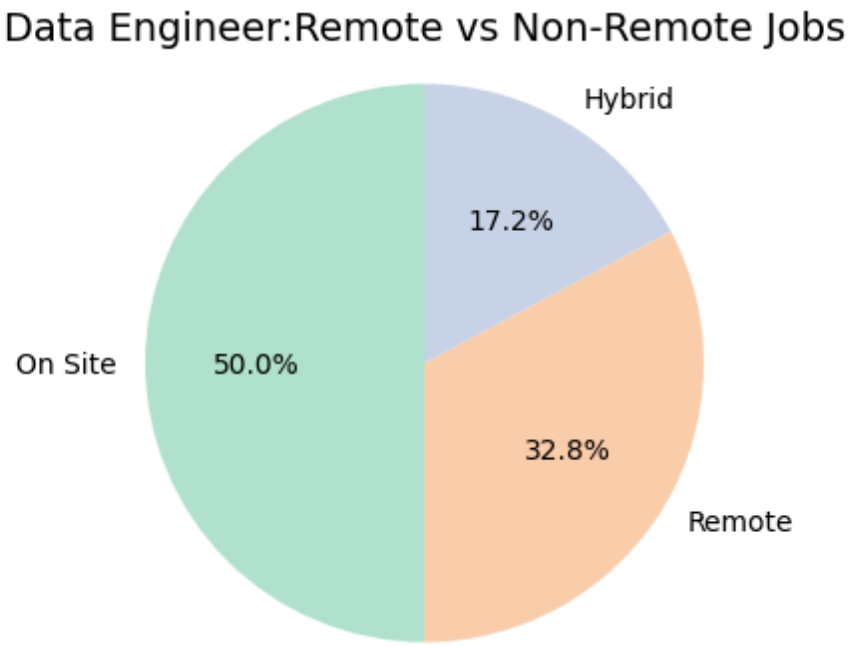
Modalidad de trabajo

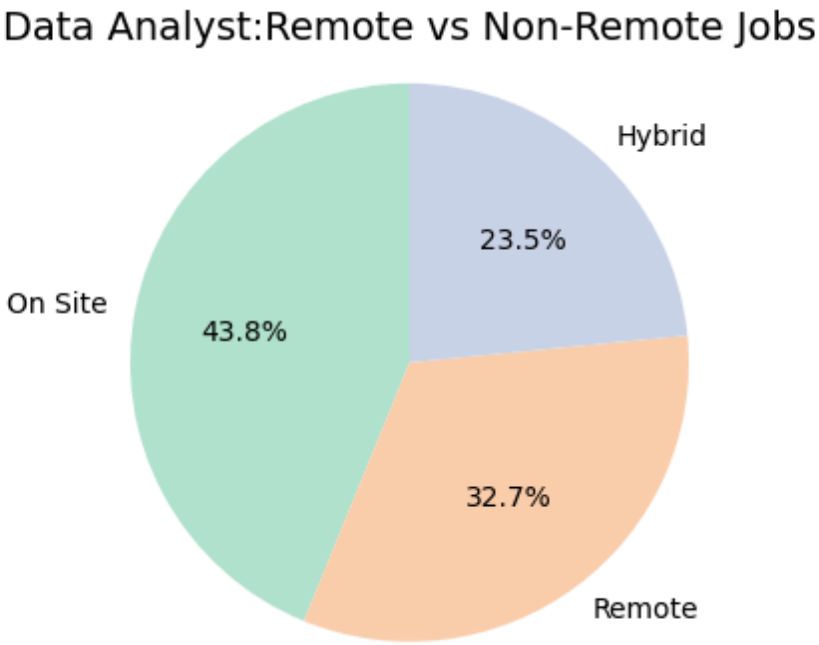
In [21]:
```
def plot_remote_distribution(df,column_name,title,category=None,figsize=(6,6)):

    def create_pie(data,chart_title,ax):
        counts = data.value_counts()
        ax.pie(
            counts,
            labels=counts.index,
            autopct='%1.1f%%',
            colors=plt.cm.Pastel2.colors,
            startangle=90
        )
        ax.set_title(chart_title,fontsize=14)
        ax.axis('equal')

    if category:
        unique_categories = df[category].unique()
        n_categories = len(unique_categories)

        fig,axes = plt.subplots(n_categories,1,figsize=figsize)
        if n_categories == 1:
            axes = [axes]

        for i, job_group in enumerate(unique_categories):
            df_grouped = df[df[category] == job_group]
            title_grouped = f"{job_group}:{title}"
            create_pie(df_grouped[column_name], title_grouped, axes[i])

        plt.tight_layout()
        plt.show()

    else:

        fig, ax = plt.subplots(figsize=figsize)
        create_pie(df[column_name],title,ax)
        plt.show()

plot_remote_distribution(df=df,column_name='workplace',title='Remote vs Non-Remo
```

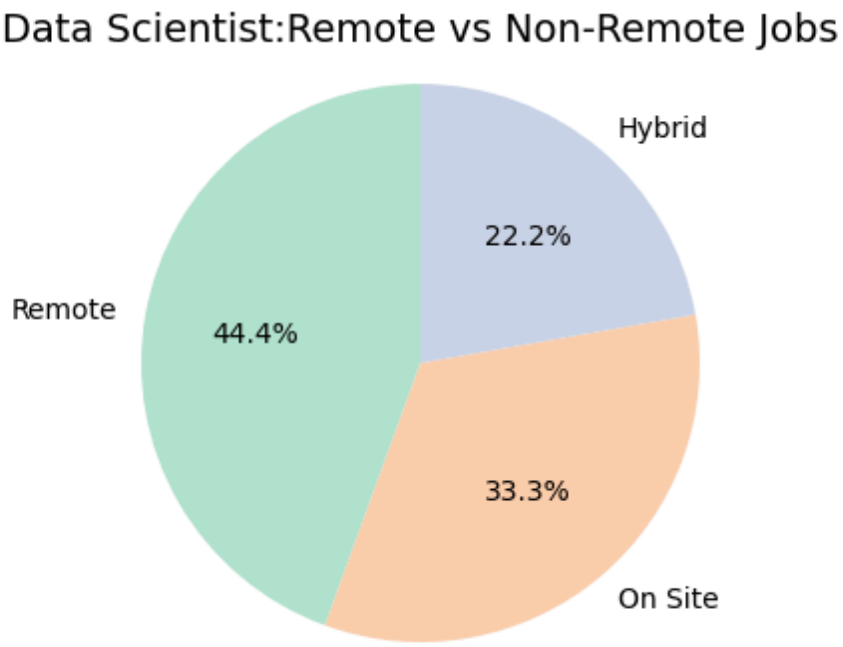


In [22]:
```
plot_remote_distribution(df=df,column_name='workplace',title='Remote vs Non-Remo
```


Business Intelligence:Remote vs Non-Remote Jobs
Hybrid
14.9%
On Site
49.4%
35.6%
Remote


Data Engineer:Remote vs Non-Remote Jobs
Hybrid
17.2%
On Site
50.0%
32.8%
Remote


Data Analyst:Remote vs Non-Remote Jobs
Hybrid
23.5%
On Site
43.8%
32.7%
Remote


Data Scientist:Remote vs Non-Remote Jobs
Hybrid
22.2%
Remote
44.4%
33.3%
On Site